

UDAY KIRAN GUDLA

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Bangalore

Targeting to leverage expertise in Embedded C, microcontroller architectures, and communication protocols to design high-reliability embedded solutions while contributing to system architecture, product innovation, and technical leadership in advanced embedded platforms.

Core Competencies

- AUTOSAR MCAL Driver Development
- Automotive Safety Standards (ISO 26262 & ASPICE)
- Embedded Firmware Development & Architecture
- Microcontroller Platforms (ARM Cortex & Aurix TriCore)
- Embedded Communication Protocols
- Device Driver Development & Hardware Abstraction
- IoT Firmware & Connected Device Development
- Hardware Debugging & Board Bring-up
- CI/CD & Embedded DevOps (Jenkins, Bitbucket, Jira)
- Cross-Functional Collaboration & Technical Leadership

Technical Skills

- Programming Languages:** Embedded C, Low-Level Assembly, C#, Python
- Microcontrollers & Platforms:** ARM Cortex-M0/M3/M33 (NXP, STMicroelectronics, Silicon Labs), Infineon Aurix TriCore
- Operating Systems:** Micrium OS, Bare-Metal Systems
- Communication Protocols & Interfaces:** I2C, SPI, UART, Ethernet, ZigBee

Profile Summary

- Embedded Systems Specialist** with 12+ years of experience designing and delivering high-reliability firmware solutions across **automotive, IoT, industrial, and railway signaling domains**, with deep expertise in **Embedded C, low-level programming, and microcontroller-based systems**.
- Automotive Software Expert** with strong hands-on experience in **AUTOSAR MCAL development for Infineon Aurix platforms**, including ownership of **Ethernet and SPI modules**, while adhering to **ASPICE processes and ISO 26262 functional safety standards**.
- Proven track record of building **connected device ecosystems and smart IoT products**, including **ZigBee/BLE enabled consumer devices**, integrating communication stacks, device drivers, and hardware interfaces for scalable product deployments.
- Highly skilled in **hardware-software integration, board bring-up, and low-level debugging**, leveraging tools such as **JTAG/SWD, logic analyzers, oscilloscopes, and GDB** to resolve complex system-level issues and optimize firmware performance.
- Experienced in **end-to-end product development within agile environments**, collaborating with global cross-functional teams while enabling **automation and release efficiency through CI/CD pipelines, scripting, and modern development toolchains**.

Career Timeline



Work Experience

Jan 2024 – Present | Staff Specialist – Automotive MCAL
Infineon Technologies | Bangalore

- Leading development and optimization of AUTOSAR MCAL drivers for Infineon Aurix 3rd Generation TriCore microcontrollers, enabling reliable and high-performance ECU software platforms for automotive applications.
- Owning the Ethernet MCAL module, driving design, development, integration, and validation activities while ensuring compliance with AUTOSAR architecture and automotive quality standards.
- Engineering SPI and Ethernet MCAL components, strengthening communication stack performance and enabling seamless integration with upper-layer AUTOSAR BSW modules.
- Ensuring adherence to automotive development frameworks, actively implementing ASPICE processes and ISO 26262 functional safety practices while maintaining high-quality documentation and traceability.
- Utilizing advanced automotive development toolchains including Jama, Enterprise Architect, Tresos, UDE, Polyspace, and Lint to support model-based development, static analysis, and compliance validation.
- Managing source control, build automation, and release workflows using Jenkins, Bitbucket, and Jira, enabling streamlined CI/CD pipelines and faster development cycles.
- Developing and maintaining Python-based automation scripts to improve tooling efficiency, build processes, and development productivity.

Debugging & Development

Tools:

JTAG/SWD, GDB, UDE,
Oscilloscopes, Logic Analyzers

Development & DevOps Tools:

Bitbucket, Jenkins, Jira, CI/CD
Pipelines

Standards & Domains:

AUTOSAR, ISO 26262, ZigBee,
Matter, Indian Railways
Signalling &
Telecommunications (S&T)

Product & Domain Experience:

Automotive MCAL
Development, IoT Devices
(Consumer & Industrial),
Railway Signalling Systems

- ⌚ Delivering multiple production-grade MCAL releases, ensuring robustness, scalability, and integration readiness for automotive software platforms.
- ⌚ Driving module-level technical ownership and cross-team collaboration, coordinating with architecture, validation, and integration teams to resolve complex system-level issues.
- ⌚ Enhancing development efficiency through automation and process improvements, reducing manual effort and strengthening reliability across release and testing pipelines.

Development Engineer – Embedded | Oct 2020 – Dec 2023

Signify (formerly Philips Lighting) | Bangalore

- ⌚ Contributed within an Agile Scrum team (10 members) to design, develop, and enhance firmware for Philips Hue smart lighting systems, enabling high-fidelity color reproduction and seamless smart-home connectivity.
- ⌚ Developed and integrated features on ZigBee application stacks from Silicon Labs, Atmel, and TI, enabling reliable wireless communication for large-scale IoT lighting ecosystems.
- ⌚ Implemented and optimized firmware modules in Embedded C, while leveraging C# and Python-based tools/frameworks to support development workflows and feature integration.
- ⌚ Designed and enhanced low-level device drivers including I2C, PWM, DMA, and UART, improving hardware abstraction and system stability across embedded platforms.
- ⌚ Worked with advanced IoT microcontrollers such as EFR32MG21, MG13, and Atmel SAMR21, supporting feature development, firmware validation, and platform integration.
- ⌚ Maintained development pipelines and release processes using Bitbucket, Jenkins, and Jira, ensuring smooth CI/CD integration and continuous feature delivery.
- ⌚ Analyzed hardware–firmware interactions using logic analyzers and oscilloscopes (CRO) to troubleshoot issues, validate communication protocols, and improve system reliability.
- ⌚ Supported multiple firmware feature releases to production, ensuring stable deployments across connected smart lighting devices.

Embedded Engineer | Apr 2017 – Oct 2020

IKA India Private Limited | Bangalore, India

- ⌚ Contributed as a key developer in a lean engineering team (2 members), driving firmware and UI development for multiple laboratory automation products including IKA Matrix, IKA T25, and IKA MultiDrive.
- ⌚ Developed embedded firmware for STM32 microcontrollers using Embedded C on Keil IDE, supporting device control, performance optimization, and reliable system operation.
- ⌚ Designed and implemented graphical user interfaces on embedded LCD displays, utilizing frameworks such as EasyGUI and TouchGfx to enhance usability and product interaction.
- ⌚ Developed and maintained device drivers for third-party BLE and Wi-Fi communication chipsets over UART, enabling wireless connectivity and external device integration.
- ⌚ Managed version control and project collaboration workflows using SVN, Jira, and Confluence, supporting documentation, issue tracking, and release coordination.
- ⌚ Collaborated with global cross-functional teams including hardware, product engineering, and international stakeholders to ensure alignment on product requirements and technical deliverables.

Previous Experience

Jul 2016 – Apr 2017 | Senior Software Engineer

Tech Mahindra (Medical Devices Team) | Bangalore

Jun 2013 – Jun 2016 | Firmware Engineer

Efftronics Systems Private Limited | Vijayawada

Education

- ⌚ **Bachelor of Technology (B.Tech)**
– **Electronics & Communication Engineering**
Vignan's Foundation for Science,
Technology & Research, Guntur
June 2009 – June 2013

Personal Details

Languages Known : English, Telugu, Hindi

Date of Birth :- 12-02-1992